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Profile

Currently working as a data scientist intern with experience in database and causal inference at Didi and previously a data science researcher with experience in recommendation systems at Microsoft. Skilled in applying machine learning (ML) techniques—such as regression analysis, ANOVA, ranking algorithms, and causal inference—to real-world diversified issues. Strong communication skills, with presentations at research symposiums.

Education

M.S. in Statistical Science (GPA: 3.8) | Duke University, Durham, NC AUG. 2025 – MAY 2027 (Expected)

- **Relevant Courses:** Machine Learning, Predictive Modeling, Python, SQL, R, Bayesian Statistics, Theory of Inference

B.A. in Mathematics & Honors Economics (GPA: 3.8) | University of Washington, Seattle, WA SEP. 2022 – JUN. 2025

- **Relevant Courses:** Data Science Computing, Econometrics, Casual Inference, Probability Theory, Nonlinear Optimization, Numerical Analysis

Experience

DATA SCIENTIST INTERN | BEIJING DIDI CHUXING TECHNOLOGY CO., LTD MAY. 2026 – CURRENT

- Supported ad-hoc requests from operations and product teams by using SQL to extract specific user groups or metrics data from the **Hive data warehouse**; assisted data analysts in providing data support for various in-depth analysis topics.
- Currently working on feasibility study of weather forecast measures on **pricing strategy** with focus on **exploring error between predicted and actual values** of multiple weather forecast features

UNDERGRADUATE RESEARCHER | MICROSOFT & DEPT OF MATHEMATICS, UW-SEATTLE MAR. – JUN. 2025

- Optimizing **large-scale network ranking algorithm** in AI Recommendation Systems through **random projections**
- **Worked with Microsoft employees** on **developing evaluation** for **computation speed** and **pairwise distance preservation**
- Applied Microsoft's ranking algorithms to evaluate ranking efficiency and stability on real-world datasets with **Python** packages **NumPy** and **Sklearn**
- Dropped **computational complexity** of the ranking algorithms by a factor of $O(1/s)$ through applying Rademacher random projection with matrix sparsity to real-world datasets
- Found **cosine similarity ranking algorithm** to be 10% stabler than dot product algorithms through computation of flipped ranking probability for both normal and Rademacher distribution random projections
- **Presented poster** at the WXML Research Symposium, tailoring responses to both technical and general audiences

UNDERGRADUATE RESEARCHER | DEPARTMENT OF ECONOMICS, UW-SEATTLE SEP. 2024 – MAR. 2025

- Evaluated the impact of benefits of IDNYC on social determinants of health of immigrant children
- Applied **data filtration**, **variable selection**, and **panel reshaping functions** `pivot_longer()`, `coalesce()` through R package `Dplyr`
- Conducted a quasi-experimental casual design study using the **Event Study Model** with **inverse probability weights**
- Paired samples with **optimal full matching techniques** with SMD less than 0.05
- **Nominated by advisor for feature spotlight** in UW newspaper and social feeds (1 of 5 students campus-wide; singular researcher).

Skills

- **Technical:** Python, PyTorch, TensorFlow, Scikit-learn, SQL, git, R, MS Office Suite
- **Leadership:** communications, team leadership, presentation
- **Languages:** English (fluent), Chinese (native), Japanese (intermediate)